JESLY PROSPER

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British citizen (unrestricted right to work in the UK)

TECHNICAL SKILLS

Programming & Data: Python (Flask, Pandas, NumPy, Scikit-learn, PyTorch), Java, JavaScript/TypeScript (React, Next.js), SQL (T-SQL, PostgreSQL, MS SQL Server), Supabase
Machine Learning & Analytics: Fraud detection, imbalanced classification (precision, recall, PR-AUC), predictive modelling (Logistic Regression, Gradient Boosting), clustering (K-Means), deep learning (CNNs, PyTorch), statistical analysis, ETL pipeline design, LLM agents (Claude API, tool use, structured output, guardrails, prompt-injection testing, LLM evaluation)
Business & Financial Analysis: Financial reporting, revenue and cost analysis, KPI tracking, forecasting support, requirements gathering, business process mapping
Cloud & Engineering Practice: AWS (Lambda, API Gateway, SQS, SNS, CloudWatch, ECR, S3, EC2), Terraform, Docker, Snowflake, Git, CI/CD pipelines, REST API design, automated testing, Agile delivery (Jira)
Reporting & AI Tools: Tableau, Power BI, advanced Excel (pivot tables, complex formulas), PowerPoint, Power Apps and Power Automate, everyday use of AI tools (Claude, GitHub Copilot) for development and analysis
Languages: Fluent in English and Haitian Creole

EDUCATION

MS in Analytics (OMSA) **Expected Winter 2027**
Georgia Institute of Technology | Computational Data Analytics track

BEng in Computer Science, 2:1 Honours **June 2026**
University of York, United Kingdom | Award mark 63. Accredited by the IET and BCS. Relevant coursework: Data Science & Machine Learning, Autonomous Robotics, Software & Systems Engineering, Object Oriented Data Structures & Algorithms

WORK EXPERIENCE

Software Developer, D&K Car Rentals LTD **June 2023 to Present**
Grand Turk, Turks and Caicos Islands | github.com/JeslyP/dk-car-rentals

- Designed, built, and solely maintained a full-stack booking, fleet, and financial reporting system (Python/Flask + SQL Server, later migrated to Next.js/Supabase) for a live commercial business.
- Analysed booking, revenue, and cost data to identify trends and inefficiencies, translating findings into recommendations that directly informed pricing and fleet decisions.
- Automated monthly financial reporting and recurring processes, replacing manual work and improving accuracy, consistency, and turnaround.
- Gathered requirements directly from non-technical stakeholders and delivered working solutions end to end, owning the system's full lifecycle from design to production support.

Freelance Software Developer, Self-Employed **March 2020 to Present**
Grand Turk, Turks and Caicos Islands / remote

- Built automated data pipelines, reporting dashboards, and web applications for multiple client businesses, integrating APIs, databases, and flat files into reliable systems.
- Implemented data validation and error-handling frameworks ensuring the accuracy and integrity of every client deliverable.

CERTIFICATIONS

Salesforce Essential Training, LinkedIn Learning **Oct 2026**
Introduction to Business Analysis, LinkedIn Learning (IIBA-accredited) **Aug 2026**
Financial Planning and Analysis (FP&A) Foundations, LinkedIn Learning **Aug 2026**
CFI Corporate Finance Foundations Professional Certificate, LinkedIn Learning **Aug 2026**
Cybersecurity Job Simulation, Mastercard (Forage) **Feb 2025**
The Legend of Python, Codédex **Feb 2025**

PROJECTS

LLM Fraud Analyst Agent, Claude with Tool Use (Personal Project) **November 2026**
github.com/JeslyP/credit-card-fraud-detection | Case study

- Built an agent in which Claude investigates a flagged transaction through 5 tools (transaction details, model score, exact score explanation, similar past cases, review policy) and drafts a structured, plain-English recommendation that a human analyst approves; a guardrail in code re-applies the written policy and customer notes are treated as untrusted.
- Evaluated it on 46 labelled cases: 100% valid structured output, 100% of model figures quoted correctly, 98% agreement with policy and 6 of 6 prompt-injection attacks resisted, at about 5 seconds per case. The evaluation exposed a rounded threshold that led the agent to clear a real fraud; the guardrail caught it, and I fixed the tool and added a regression test. CI logs in to Claude with short-lived federated tokens (Workload Identity Federation) instead of a stored API key.

Fraud Detection: Models, Behavioural Features and Scoring Service (Personal Project) **October 2026**
github.com/JeslyP/credit-card-fraud-detection | Case study | Live API

- Built class-weighted logistic regression and gradient boosting models in scikit-learn on 284,807 card transactions (0.17% fraud), using a time-based split so the model was tested only on later transactions it had never seen.
- Caught 85% of fraud (63 of 74) while flagging just 0.5% of genuine transactions (PR-AUC 0.746); set the alert threshold by the cost of missed fraud versus manual review, cutting estimated fraud losses by 51%, and tracked recall over time to detect drift.
- On 6.3M PaySim payments, built behavioural features from each account's past activity (payee history, 24-hour velocity, new payees), which tripled PR-AUC from 0.14 to 0.41; excluded balance columns that leak the label.
- Deployed it live on AWS with Terraform from a one-click GitHub Actions pipeline: a FastAPI scoring API on Lambda and API Gateway, an event-driven SQS consumer publishing fraud alerts to SNS with a dead-letter queue, CloudWatch alarms and automated tests in CI.

Autonomous Robotics System (University of York) **January 2026**
github.com/JeslyP/AUROJES2025

- Designed and implemented an autonomous multi-robot barrel-collection system for TurtleBot3 Waffle Pi robots in a Gazebo simulation, using Python and ROS2 within a Dockerised development environment.
- Built an 8-state finite state machine (searching, approaching, positioning, picking up, delivering, offloading, clearing space, decontaminating) to coordinate patrol navigation, target selection, pickup, delivery, and decontamination behaviours.
- Used the Nav2 stack for waypoint-based patrol navigation and LiDAR-based obstacle sectoring for real-time obstacle avoidance, combined with a proportional controller for visual servoing to align and approach detected targets.
- Implemented hysteresis-based target selection to prevent oscillation between similarly-sized detections, plus a timeout and escape manoeuvre to recover automatically when the robot became stuck.
- Designed and ran five distinct test scenarios (baseline, sensor noise robustness, varied item distribution via random seed, open environment, and a full-complexity stress test combining noise and obstacles), each run for 3 hours of simulation time, achieving 90 to 100% detection accuracy across scenarios.

NHS Data Science & ETL Pipeline (University of York) **May 2025**
github.com/JeslyP/NHS-Data-Science-Project-

- Designed a normalised (3NF) SQLite database with an ER diagram from six raw NHS operational datasets (appointments, billing, insurance, prescriptions, surgery, test records), covering 557k appointments and 400k patients.
- Tested what drives missed appointments with chi-square tests and a logistic regression model, and reported honestly that no factor predicted cancellation (ROC-AUC 0.50); used K-means clustering to find that 2% of patients account for 15% of appointments.

CIFAR-10 Image Classification (University of York) **May 2025**
github.com/JeslyP/IMLOCW

- Built a CNN from scratch in PyTorch (three convolutional blocks with batch normalisation and dropout, trained with Adam and a scheduled learning rate over 20 epochs), achieving 79% test accuracy on a 40,000/10,000/10,000 train/validation/test split.

Campus Tycoon, Java Simulation Game (University of York) **December 2024**
github.com/JeslyP/CampusTycoonBackend2

- Built a libGDX-based city simulation game in Java, using the Factory pattern to instantiate building types and inheritance/polymorphism across building and event class hierarchies, coordinating a 4-person Agile team with Jira and delivering on time with CI/CD integration.
- Diagnosed why the project's JUnit tests had never run (wrong source set), fixed an OpenGL initialisation bug they exposed, and added a GitHub Actions pipeline that builds and runs all tests on every push.